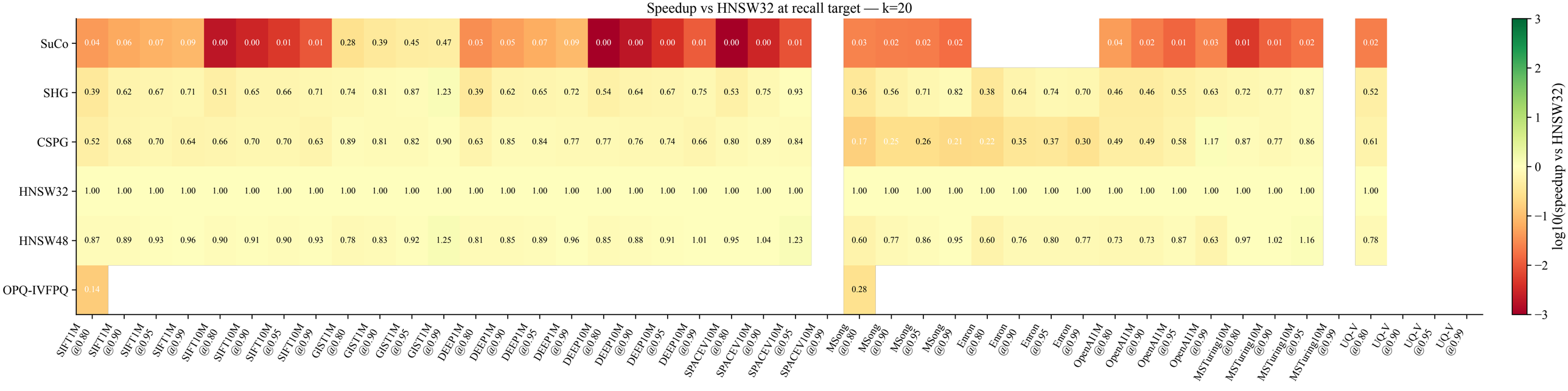
Speedup vs HNSW32 at recall target — k=20
SuCo
SHG
CSPG
HNSW32
HNSW48
OPQ-IVFPQ
0.04 0.06 0.07 0.09 0.00 0.00 0.01 0.01 0.28 0.39 0.45 0.47 0.03 0.05 0.07 0.09 0.00 0.00 0.00 0.01 0.00 0.00 0.01 0.03 0.02 0.02 0.02 0.04 0.02 0.01 0.03 0.01 0.01 0.02 0.02
0.39 0.62 0.67 0.71 0.51 0.65 0.66 0.71 0.74 0.81 0.87 1.23 0.39 0.62 0.65 0.72 0.54 0.64 0.67 0.75 0.53 0.75 0.93 0.36 0.56 0.71 0.82 0.38 0.64 0.74 0.70 0.46 0.46 0.55 0.63 0.72 0.77 0.87 0.52
0.52 0.68 0.70 0.64 0.66 0.70 0.70 0.63 0.89 0.81 0.82 0.90 0.63 0.85 0.84 0.77 0.77 0.76 0.74 0.66 0.80 0.89 0.84 0.17 0.25 0.26 0.21 0.22 0.35 0.37 0.30 0.49 0.49 0.58 1.17 0.87 0.77 0.86 0.61
1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00
0.87 0.89 0.93 0.96 0.90 0.91 0.90 0.93 0.78 0.83 0.92 1.25 0.81 0.85 0.89 0.96 0.85 0.88 0.91 1.01 0.95 1.04 1.23 0.60 0.77 0.86 0.95 0.60 0.76 0.80 0.77 0.73 0.73 0.87 0.63 0.97 1.02 1.16 0.78
0.14 0.28
SIFT1M @0.80
SIFT1M @0.90
SIFT1M @0.95
SIFT1M @0.99
SIFT10M @0.80
SIFT10M @0.90
SIFT10M @0.95
SIFT10M @0.99
GIST1M @0.80
GIST1M @0.90
GIST1M @0.95
GIST1M @0.99
DEEP1M @0.80
DEEP1M @0.90
DEEP1M @0.95
DEEP1M @0.99
DEEP10M @0.80
DEEP10M @0.90
DEEP10M @0.95
DEEP10M @0.99
SPACEV10M @0.80
SPACEV10M @0.90
SPACEV10M @0.95
SPACEV10M @0.99
MSong @0.80
MSong @0.90
MSong @0.95
MSong @0.99
Enron @0.80
Enron @0.90
Enron @0.95
Enron @0.99
OpenAI1M @0.80
OpenAI1M @0.90
OpenAI1M @0.95
OpenAI1M @0.99
MSTuring10M @0.80
MSTuring10M @0.90
MSTuring10M @0.95
MSTuring10M @0.99
UQ-V @0.80
UQ-V @0.90
UQ-V @0.95
UQ-V @0.99
log10(speedup vs HNSW32)
3
2
1
0
−1
−2
−3